

HAFEEZA ZULFIQAR

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AI/ML Engineer focused on Generative AI, Computer Vision, and RAG systems, building end-to-end solutions for automation and conversational intelligence.

Education

Mohammad Ali Jinnah University Bachelors in Software Engineering	Karachi, Pakistan Feb2020 - Feb2024
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Professional Experience

Proxima AI <i>AI Engineer</i> Delivering cutting-edge machine learning and NLP-based solutions.	Karachi, Pakistan Mar2024 - Present
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- Developed **end-to-end AI-powered multimedia analytics pipelines** for processing live broadcast streams.
- Built Speech-to-Text pipelines using **Whisper ASR, Voice Activity Detection (VAD), speaker diarization, and speaker identification**.
- Developed OCR systems for **handwritten, printed, and Urdu text extraction** with high accuracy.
- Implemented NLP pipelines including **summarization, sentiment analysis, Named Entity Recognition (NER), and text classification**.
- Worked on RAG-based conversational **AI systems using embeddings, vector databases, LangChain, and LLM-powered retrieval workflows**.
- Fine-tuned LLMs and OCR models** for domain-specific AI applications and automation tasks.
- Built scalable backend APIs and AI workflows** using Python, FastAPI, and modern deep learning frameworks.
- Developed **web scraping and automation pipelines for collecting structured and multimedia data** from platforms including YouTube, LinkedIn, and Google Ads.
- Designed AI-driven automation systems including **conversational bots and real-time intelligent processing pipelines**.

Projects

AI Broadcast Intelligence System — <i>AI Engineer Generative AI, RAG, NLP, Computer Vision</i> An AI-powered platform for analyzing live news broadcasts.	Jan 2026 – Present
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- Built pipelines for Live Stream Recording, OCR Headlines Extraction, and Speech-to-Text Transcription.
- Implemented VAD, Whisper ASR, Speaker Diarization, Speaker Identification, and Face Detection for multilingual news processing.
- Developed NLP workflows for Summarization, NER, and Sentiment Analysis.
- Created RAG-based Conversational AI using Embeddings, Vector DBs, LangChain, and LLM Q&A.
- Enabled semantic search over news transcripts and headlines.

Medical Slip Duplicate Detection System — <i>AI/Computer Vision Engineer</i> An AI system for detecting duplicate medical slips and reports.	Jan 2025 – Mar 2025
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- Built models to classify Medical Slips, X-rays, and other healthcare documents.
- Added Handwritten vs Printed document classification before OCR.
- Implemented OCR-based and visual similarity detection using Edge Detection and Background Matching.
- Combined text and image analysis to generate duplicate similarity reports.

Smart Startup Assessment — <i>Founder & Product Lead</i> An AI-powered platform helping founders improve clarity, team alignment, and pitch readiness.	Oct 2024 – Jan 2025
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- Analyzes startup documents (problem, solution, roadmap) using NLP for structure and clarity.
- Provides targeted feedback to address weak points in startup presentations.
- Evaluates team fit by analyzing resumes using AI.

- Reduces repetitive work for judges through a multi-startup review system.
- Aims to prevent early-stage startup failure by improving communication and decision-making.

<https://github.com/HafeezaZulfiqar>

Certifications

Huggingface Certification

– LLM Course & Agents Course